

Facial Expression Recognition: A Comparative Analysis of Traditional Machine Learning and Deep Learning Approaches

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Abstract—The goal of this project is to evaluate and compare the effectiveness of traditional statistical computer vision techniques against modern deep learning architectures for the task of facial expression recognition. We will implement a manual feature extraction pipeline using Histogram of Oriented Gradients (HOG) combined with a Support Vector Machine (SVM) classifier, and compare it against a custom-built Convolutional Neural Network (CNN). Furthermore, we will benchmark our models against state-of-the-art pre-trained deep learning frameworks (FaceNet and VGG-Face) via the DeepFace library. This comparative study aims to highlight the trade-offs between computational efficiency, feature interpretability, and raw predictive accuracy in computer vision systems.

I. INTRODUCTION

Facial expression recognition (FER) is a fundamental problem in computer vision with broad applications in human-computer interaction, automated behavioral analysis, and affective computing [6]. The core challenge involves accurately classifying static images of human faces into distinct emotional categories such as happiness, sadness, anger, fear, surprise, disgust, and neutral.

The primary motivation for this project is to systematically investigate the evolutionary shift in computer vision methodologies. Historically, practitioners relied on handcrafted feature extraction methods to feed statistical classifiers. Today, deep learning models learn these feature representations automatically from raw data, often achieving superior accuracy at the cost of higher computational demand and lower interpretability [6], [8]. By building and comparing both a traditional pipeline and a deep learning model, and benchmarking them against current state-of-the-art industry models like FaceNet [9] and VGG-Face [10], we aim to quantify these trade-offs and gain a deeper understanding of both paradigms.

II. METHOD

A. Outline of Methods

The project will be executed in three primary phases, representing three distinct approaches to the problem:

- 1) **Traditional Statistical Pipeline:** We will implement a classical machine learning approach. First, images will

be preprocessed (grayscaled and normalized). We will then extract spatial and texture features using Histogram of Oriented Gradients (HOG) [3] and Local Binary Patterns (LBP) [4]. These manually engineered features will be utilized to train a Support Vector Machine (SVM) classifier [5].

- 2) **Deep Learning Pipeline:** We will design and train a custom Convolutional Neural Network (CNN). This model will take raw pixel data as input and learn feature representations through convolutional and pooling layers, concluding with connected layers for classification.
- 3) **State-of-the-Art Benchmark:** To evaluate our models against widely used deep learning approaches, we will utilize the open-source DeepFace library [7]. By leveraging pre-trained deep neural network models provided through the DeepFace framework, we establish a robust baseline for comparison

B. Datasets

The primary dataset for training and evaluation will be the FER2013 dataset [1]. It consists of 35,887 grayscale face images, each sized at 48×48 pixels, categorized into seven core emotions. If project time permits, we will also utilize the Extended Cohn-Kanade (CK+) dataset [2] to perform cross-dataset evaluation and test the robustness of our trained models on unseen data distributions.

C. Expected Results and Evaluation

We anticipate that the custom CNN will outperform the traditional HOG+SVM pipeline in terms of raw accuracy, as it can learn more complex, non-linear representations of facial features. However, we expect the SVM approach to train significantly faster and require less computational overhead. The pre-trained models (FaceNet/VGG-Face) are expected to represent the state-of-the-art upper bound.

Performance will be rigorously evaluated using overall classification accuracy, precision, recall, and F1-scores. We will also utilize confusion matrices to identify specific emotional misclassifications.

III. PROJECT SCHEDULE

The timeline below outlines the subtasks and responsibilities to ensure the project meets the deadlines specified in the course syllabus.

TABLE I
PROJECT TIMELINE AND RESPONSIBILITIES

Subtask	Approach	Deadline	Owner
Initial Literature Search	Researching HOG/SVM and CNN structures for FER	Mar 24	Berre Delikara, Berra Eđcin
Data Preprocessing	Cleaning FER2013, implementing HOG/LBP extraction	Apr 10	Berra Eđcin, Mijgan Selen Karakař
Traditional Model	Training and tuning the SVM classifier	Apr 15	Berre Delikara, Berra Eđcin
Deep Learning Model	Designing, training, and validating custom CNN	Apr 25	Mijgan Selen Karakař, Berre Delikara
SOTA Benchmarking	Implementing DeepFace and running evaluations	May 5	Berra Eđcin, Mijgan Selen Karakař
Final Report Preparation	Documenting results and ablation studies	May 10	All
Final Presentation	Slide preparation and demo setup	May 13	All

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